

AP1_A2_TRAN Ngoc Hieu Van_22205175

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Introduction

This report analyzes historical data of FIFA World Cup using R Studio. The analysis includes data visualization, summary statistics, investigation on distribution of home penalty, home team scores, and match results across several factors, and filter out top 5 teams in penalty shootout matches.

Load all the dataset

```
matches <- read.csv("Matches.csv")
stadiums <- read.csv("Stadiums.csv")
teams <- read.csv("Teams.csv")
tournaments <- read.csv("Tournaments.csv")
```

Question 1

Inspect the data structure

View the structure of each dataset

```
str(matches)
```

```
## 'data.frame':    964 obs. of  16 variables:
##  $ MatchID      : chr  "M-1930-01" "M-1930-02" "M-1930-03" "M-1930-04" ...
##  $ TournamentID : chr  "WC-1930" "WC-1930" "WC-1930" "WC-1930" ...
##  $ MatchName     : chr  "France v Mexico" "United States v Belgium" "Yugoslavia v Brazil" "Romania v" ...
##  $ Stage         : chr  "group stage" "group stage" "group stage" "group stage" ...
##  $ MatchDate     : chr  "7/13/1930" "7/13/1930" "7/14/1930" "7/14/1930" ...
##  $ MatchTime     : chr  "15:00:00" "15:00:00" "12:45:00" "14:50:00" ...
##  $ StadiumID     : chr  "S-193" "S-192" "S-192" "S-193" ...
##  $ HomeTeamID    : chr  "T-28" "T-80" "T-84" "T-59" ...
##  $ AwayTeamID    : chr  "T-44" "T-06" "T-09" "T-54" ...
##  $ HomeTeamScore : int   4 3 2 3 1 3 4 3 1 1 ...
##  $ AwayTeamScore : int   1 0 1 1 0 0 0 0 0 0 ...
##  $ ExtraTime     : int   0 0 0 0 0 0 0 0 0 0 ...
##  $ PenaltyShootout: int   0 0 0 0 0 0 0 0 0 0 ...
##  $ HomePenalty   : chr   "?" "0" "0" "0" ...
##  $ AwayPenalty   : chr   "0" "0" "0" "?" ...
##  $ Result        : chr  "home team win" "home team win" "home team win" "home team win" ...
```

```
str(stadiums)
```

```
## 'data.frame': 193 obs. of 4 variables:
## $ StadiumID : chr "S-089" "S-108" "S-107" "S-111" ...
## $ StadiumName: chr "Jita Stadium" "Ahmad bin Ali Stadium" "Al Bayt Stadium" "Al Janoub Stadium" ..
## $ City : chr "Jita" "Al Rayyan" "Al Khor" "Al Wakrah" ...
## $ Country : chr "Japan" "Qatar" "Qatar" "Qatar" ...
```

```
str(teams)
```

```
## 'data.frame': 85 obs. of 3 variables:
## $ TeamID : chr "T-01" "T-02" "T-03" "T-04" ...
## $ TeamName: chr "Algeria" "Angola" "Argentina" "Australia" ...
## $ TeamCode: chr "DZA" "AGO" "ARG" "AUS" ...
```

```
str(tournaments)
```

```
## 'data.frame': 23 obs. of 2 variables:
## $ TournamentID : chr "WC-1930" "WC-1934" "WC-1938" "WC-1950" ...
## $ TournamentName: chr "1930 FIFA World Cup" "1934 FIFA World Cup" "1938 FIFA World Cup" "1950 FIFA
```

Display the first six observations to quickly view the layout of each dataset

```
head(matches)
```

```
##      MatchID TournamentID      MatchName      Stage MatchDate
## 1 M-1930-01      WC-1930      France v Mexico group stage 7/13/1930
## 2 M-1930-02      WC-1930 United States v Belgium group stage 7/13/1930
## 3 M-1930-03      WC-1930      Yugoslavia v Brazil group stage 7/14/1930
## 4 M-1930-04      WC-1930      Romania v Peru group stage 7/14/1930
## 5 M-1930-05      WC-1930      Argentina v France group stage 7/15/1930
## 6 M-1930-06      WC-1930      Chile v Mexico group stage 7/16/1930
##      MatchTime StadiumID HomeTeamID AwayTeamID HomeTeamScore AwayTeamScore
## 1 15:00:00      S-193      T-28      T-44      4      1
## 2 15:00:00      S-192      T-80      T-06      3      0
## 3 12:45:00      S-192      T-84      T-09      2      1
## 4 14:50:00      S-193      T-59      T-54      3      1
## 5 16:00:00      S-192      T-03      T-28      1      0
## 6 14:45:00      S-192      T-13      T-44      3      0
##      ExtraTime PenaltyShootout HomePenalty AwayPenalty      Result
## 1      0      0      ?      0 home team win
## 2      0      0      0      0 home team win
## 3      0      0      0      0 home team win
## 4      0      0      0      ? home team win
## 5      0      0      0      0 home team win
## 6      0      0      0      0 home team win
```

```
head(stadiums)
```

```
## StadiumID StadiumName City Country
## 1 S-089 Jita Stadium Jita Japan
## 2 S-108 Ahmad bin Ali Stadium Al Rayyan Qatar
## 3 S-107 Al Bayt Stadium Al Khor Qatar
## 4 S-111 Al Janoub Stadium Al Wakrah Qatar
## 5 S-112 Al Thumama Stadium Doha Qatar
## 6 S-064 Allianz Arena Munich Germany
```

```
head(teams)
```

```
## TeamID TeamName TeamCode
## 1 T-01 Algeria DZA
## 2 T-02 Angola AGO
## 3 T-03 Argentina ARG
## 4 T-04 Australia AUS
## 5 T-05 Austria AUT
## 6 T-06 Belgium BEL
```

```
head(tournaments)
```

```
## TournamentID TournamentName
## 1 WC-1930 1930 FIFA World Cup
## 2 WC-1934 1934 FIFA World Cup
## 3 WC-1938 1938 FIFA World Cup
## 4 WC-1950 1950 FIFA World Cup
## 5 WC-1954 1954 FIFA World Cup
## 6 WC-1958 1958 FIFA World Cup
```

Run summary to present the class and statistical distribution of every column in each dataset

```
summary(matches)
```

```
## MatchID TournamentID MatchName Stage
## Length:964 Length:964 Length:964 Length:964
## Class :character Class :character Class :character Class :character
## Mode :character Mode :character Mode :character Mode :character
##
##
## MatchDate MatchTime StadiumID HomeTeamID
## Length:964 Length:964 Length:964 Length:964
## Class :character Class :character Class :character Class :character
## Mode :character Mode :character Mode :character Mode :character
##
##
## AwayTeamID HomeTeamScore AwayTeamScore ExtraTime
## Length:964 Min. : 0.000 Min. :0.000 Min. :0.00000
## Class :character 1st Qu.: 1.000 1st Qu.:0.000 1st Qu.:0.00000
## Mode :character Median : 1.000 Median :1.000 Median :0.00000
## Mean : 1.767 Mean :1.055 Mean :0.07573
## 3rd Qu.: 3.000 3rd Qu.:2.000 3rd Qu.:0.00000
```

```
##           Max.    :10.000   Max.    :7.000   Max.    :1.00000
## PenaltyShootout HomePenalty   AwayPenalty      Result
## Min.    :0.00000   Length:964   Length:964   Length:964
## 1st Qu.:0.00000   Class :character Class :character Class :character
## Median :0.00000   Mode  :character Mode  :character Mode  :character
## Mean    :0.03631
## 3rd Qu.:0.00000
## Max.    :1.00000
```

```
summary(stadiums)
```

```
##   StadiumID      StadiumName      City      Country
## Length:193      Length:193      Length:193 Length:193
## Class :character Class :character Class :character Class :character
## Mode  :character Mode  :character Mode  :character Mode  :character
```

```
summary(teams)
```

```
##   TeamID      TeamName      TeamCode
## Length:85      Length:85      Length:85
## Class :character Class :character Class :character
## Mode  :character Mode  :character Mode  :character
```

```
summary(tournaments)
```

```
##   TournamentID   TournamentName
## Length:23        Length:23
## Class :character Class :character
## Mode  :character Mode  :character
```

Replace the missing values as “?” with “NA”

```
matches[matches == "?"] <- NA # Replace the "?" values with "NA"
str(matches) # Check if the missing values "?" are replaced with NA
```

```
## 'data.frame':   964 obs. of  16 variables:
## $ MatchID      : chr  "M-1930-01" "M-1930-02" "M-1930-03" "M-1930-04" ...
## $ TournamentID : chr  "WC-1930" "WC-1930" "WC-1930" "WC-1930" ...
## $ MatchName    : chr  "France v Mexico" "United States v Belgium" "Yugoslavia v Brazil" "Romania v" ...
## $ Stage        : chr  "group stage" "group stage" "group stage" "group stage" ...
## $ MatchDate    : chr  "7/13/1930" "7/13/1930" "7/14/1930" "7/14/1930" ...
## $ MatchTime    : chr  "15:00:00" "15:00:00" "12:45:00" "14:50:00" ...
## $ StadiumID    : chr  "S-193" "S-192" "S-192" "S-193" ...
## $ HomeTeamID   : chr  "T-28" "T-80" "T-84" "T-59" ...
## $ AwayTeamID   : chr  "T-44" "T-06" "T-09" "T-54" ...
## $ HomeTeamScore : int  4 3 2 3 1 3 4 3 1 1 ...
## $ AwayTeamScore : int  1 0 1 1 0 0 0 0 0 0 ...
## $ ExtraTime    : int  0 0 0 0 0 0 0 0 0 0 ...
## $ PenaltyShootout: int  0 0 0 0 0 0 0 0 0 0 ...
## $ HomePenalty   : chr  NA "0" "0" "0" ...
## $ AwayPenalty   : chr  "0" "0" "0" NA ...
## $ Result        : chr  "home team win" "home team win" "home team win" "home team win" ...
```

Convert categorical variables to factors

```
matches$Result <- as.factor(matches$Result)
matches$Stage <- as.factor(matches$Stage)
stadiums$Country <- as.factor(stadiums$Country)
matches$ExtraTime <- as.factor(matches$ExtraTime)
```

Convert penalties to numeric format

```
matches$HomePenalty <- as.numeric(matches$HomePenalty)
matches$AwayPenalty <- as.numeric(matches$AwayPenalty)
```

Replace the missing value in penalties with 0

```
matches$HomePenalty[is.na(matches$HomePenalty)] <- 0
matches$AwayPenalty[is.na(matches$AwayPenalty)] <- 0
```

Validate the pre-processing results

```
colSums(is.na(matches)) # Check remaining missing values
```

```
##      MatchID      TournamentID      MatchName      Stage      MatchDate
##           0              0              0              0              0
##      MatchTime      StadiumID      HomeTeamID      AwayTeamID      HomeTeamScore
##           0              0              0              0              0
##      AwayTeamScore      ExtraTime      PenaltyShootout      HomePenalty      AwayPenalty
##           0              0              0              0              0
##           Result
##           0
```

```
str(matches) # Check variable types in the matches dataset
```

```
## 'data.frame':   964 obs. of  16 variables:
##  $ MatchID      : chr  "M-1930-01" "M-1930-02" "M-1930-03" "M-1930-04" ...
##  $ TournamentID : chr  "WC-1930" "WC-1930" "WC-1930" "WC-1930" ...
##  $ MatchName    : chr  "France v Mexico" "United States v Belgium" "Yugoslavia v Brazil" "Romania v" ...
##  $ Stage        : Factor w/ 8 levels "final","final round",...: 3 3 3 3 3 3 3 3 3 ...
##  $ MatchDate    : chr  "7/13/1930" "7/13/1930" "7/14/1930" "7/14/1930" ...
##  $ MatchTime    : chr  "15:00:00" "15:00:00" "12:45:00" "14:50:00" ...
##  $ StadiumID    : chr  "S-193" "S-192" "S-192" "S-193" ...
##  $ HomeTeamID   : chr  "T-28" "T-80" "T-84" "T-59" ...
##  $ AwayTeamID   : chr  "T-44" "T-06" "T-09" "T-54" ...
##  $ HomeTeamScore: int   4 3 2 3 1 3 4 3 1 1 ...
##  $ AwayTeamScore: int   1 0 1 1 0 0 0 0 0 0 ...
##  $ ExtraTime    : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 ...
##  $ PenaltyShootout: int   0 0 0 0 0 0 0 0 0 0 ...
##  $ HomePenalty  : num   0 0 0 0 0 0 0 0 0 0 ...
##  $ AwayPenalty  : num   0 0 0 0 0 0 0 0 0 0 ...
##  $ Result       : Factor w/ 3 levels "away team win",...: 3 3 3 3 3 3 3 3 3 ...
```

```
str(stadiums) # Check variable types in the stadium dataset
```

```
## 'data.frame':    193 obs. of  4 variables:
## $ StadiumID : chr  "S-089" "S-108" "S-107" "S-111" ...
## $ StadiumName: chr  "Jita Stadium" "Ahmad bin Ali Stadium" "Al Bayt Stadium" "Al Janoub Stadium" ..
## $ City       : chr  "Jita" "Al Rayyan" "Al Khor" "Al Wakrah" ...
## $ Country    : Factor w/ 18 levels "Argentina","Brazil",...: 8 10 10 10 10 6 6 2 2 2 ...
```

```
sum(is.na(matches$HomePenalty)) # Confirm no missing value in HomePenalty
```

```
## [1] 0
```

```
sum(is.na(matches$AwayPenalty)) # Confirm no missing value in AwayPenalty
```

```
## [1] 0
```

Visualize the distribution of HomePenalty for matches that involved a penalty shootout

Filter matches with a penalty shootout

```
library(dplyr) # Use the dplyr library for filtering
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
unique(matches$PenaltyShootout) # Check the value of PenaltyShootout
```

```
## [1] 0 1
```

```
penalty_matches <- matches %>%
  filter(PenaltyShootout == 1) # Filter out the penalty shootout matches
penalty_matches # Check if the penalty_matches is filtered correctly
```

```
##      MatchID TournamentID      MatchName      Stage
## 1 M-1982-50      WC-1982  West Germany v France  semi-finals
## 2 M-1986-45      WC-1986    Brazil v France  quarter-finals
## 3 M-1986-46      WC-1986  West Germany v Mexico  quarter-finals
## 4 M-1986-48      WC-1986    Spain v Belgium  quarter-finals
## 5 M-1990-41      WC-1990 Republic of Ireland v Romania  round of 16
```

## 6	M-1990-45	WC-1990	Argentina v Yugoslavia	quarter-finals		
## 7	M-1990-49	WC-1990	Argentina v Italy	semi-finals		
## 8	M-1990-50	WC-1990	West Germany v England	semi-finals		
## 9	M-1994-44	WC-1994	Mexico v Bulgaria	round of 16		
## 10	M-1994-48	WC-1994	Romania v Sweden	quarter-finals		
## 11	M-1994-52	WC-1994	Brazil v Italy	final		
## 12	M-1998-56	WC-1998	Argentina v England	round of 16		
## 13	M-1998-57	WC-1998	Italy v France	quarter-finals		
## 14	M-1998-61	WC-1998	Brazil v Netherlands	semi-finals		
## 15	M-2002-52	WC-2002	Spain v Republic of Ireland	round of 16		
## 16	M-2002-59	WC-2002	Spain v South Korea	quarter-finals		
## 17	M-2006-54	WC-2006	Switzerland v Ukraine	round of 16		
## 18	M-2006-57	WC-2006	Germany v Argentina	quarter-finals		
## 19	M-2006-59	WC-2006	England v Portugal	quarter-finals		
## 20	M-2006-64	WC-2006	Italy v France	final		
## 21	M-2010-55	WC-2010	Paraguay v Japan	round of 16		
## 22	M-2010-58	WC-2010	Uruguay v Ghana	quarter-finals		
## 23	M-2014-49	WC-2014	Brazil v Chile	round of 16		
## 24	M-2014-52	WC-2014	Costa Rica v Greece	round of 16		
## 25	M-2014-60	WC-2014	Netherlands v Costa Rica	quarter-finals		
## 26	M-2014-62	WC-2014	Netherlands v Argentina	semi-finals		
## 27	M-2018-51	WC-2018	Spain v Russia	round of 16		
## 28	M-2018-52	WC-2018	Croatia v Denmark	round of 16		
## 29	M-2018-56	WC-2018	Colombia v England	round of 16		
## 30	M-2018-60	WC-2018	Russia v Croatia	quarter-finals		
## 31	M-2022-53	WC-2022	Japan v Croatia	round of 16		
## 32	M-2022-55	WC-2022	Morocco v Spain	round of 16		
## 33	M-2022-57	WC-2022	Croatia v Brazil	quarter-finals		
## 34	M-2022-58	WC-2022	Netherlands v Argentina	quarter-finals		
## 35	M-2022-64	WC-2022	Argentina v France	final		
##	MatchDate	MatchTime	StadiumID	HomeTeamID	AwayTeamID	HomeTeamScore
## 1	07/08/1982	21:00:00	S-159	T-83	T-28	3
## 2	6/21/1986	12:00:00	S-095	T-09	T-28	1
## 3	6/21/1986	16:00:00	S-102	T-83	T-44	0
## 4	6/22/1986	16:00:00	S-104	T-71	T-06	1
## 5	6/25/1990	17:00:00	S-073	T-58	T-59	0
## 6	6/30/1990	17:00:00	S-071	T-03	T-84	0
## 7	07/03/1990	20:00:00	S-076	T-03	T-39	1
## 8	07/04/1990	20:00:00	S-082	T-83	T-27	1
## 9	07/05/1994	16:30:00	S-184	T-44	T-10	1
## 10	07/10/1994	12:30:00	S-189	T-59	T-72	2
## 11	7/17/1994	12:30:00	S-187	T-09	T-39	0
## 12	6/30/1998	21:00:00	S-049	T-03	T-27	2
## 13	07/03/1998	16:30:00	S-048	T-39	T-28	0
## 14	07/07/1998	21:00:00	S-042	T-09	T-46	1
## 15	6/16/2002	20:30:00	S-145	T-71	T-58	1
## 16	6/22/2002	15:30:00	S-140	T-71	T-69	0
## 17	6/26/2006	21:00:00	S-054	T-73	T-78	0
## 18	6/30/2006	17:00:00	S-053	T-29	T-03	1
## 19	07/01/2006	17:00:00	S-058	T-27	T-56	0
## 20	07/09/2006	20:00:00	S-053	T-39	T-28	1
## 21	6/29/2010	16:00:00	S-135	T-53	T-42	0
## 22	07/02/2010	20:30:00	S-131	T-81	T-30	1
## 23	6/28/2014	13:00:00	S-008	T-09	T-13	1

## 24	6/29/2014	17:00:00	S-018	T-16	T-31	1
## 25	07/05/2014	17:00:00	S-021	T-46	T-16	0
## 26	07/09/2014	17:00:00	S-022	T-46	T-03	0
## 27	07/01/2018	17:00:00	S-117	T-71	T-60	1
## 28	07/01/2018	21:00:00	S-119	T-17	T-21	1
## 29	07/03/2018	21:00:00	S-118	T-15	T-27	1
## 30	07/07/2018	21:00:00	S-124	T-60	T-17	2
## 31	12/05/2022	18:00:00	S-111	T-42	T-17	1
## 32	12/06/2022	18:00:00	S-109	T-45	T-71	0
## 33	12/09/2022	18:00:00	S-109	T-17	T-09	1
## 34	12/09/2022	22:00:00	S-114	T-46	T-03	2
## 35	12/18/2022	18:00:00	S-114	T-03	T-28	3

##	AwayTeamScore	ExtraTime	PenaltyShootout	HomePenalty	AwayPenalty
## 1	3	1	1	0	4
## 2	1	1	1	0	4
## 3	0	1	1	0	1
## 4	1	1	1	0	5
## 5	0	1	1	5	4
## 6	0	1	1	3	2
## 7	1	1	1	4	3
## 8	1	1	1	4	3
## 9	1	1	1	1	3
## 10	2	1	1	4	5
## 11	0	1	1	3	2
## 12	2	1	1	4	3
## 13	0	1	1	3	4
## 14	1	1	1	4	2
## 15	1	1	1	3	2
## 16	0	1	1	3	5
## 17	0	1	1	0	3
## 18	1	1	1	4	2
## 19	0	1	1	1	3
## 20	1	1	1	5	3
## 21	0	1	1	5	3
## 22	1	1	1	4	2
## 23	1	1	1	3	2
## 24	1	1	1	5	3
## 25	0	1	1	4	3
## 26	0	1	1	2	4
## 27	1	1	1	3	4
## 28	1	1	1	3	2
## 29	1	1	1	3	4
## 30	2	1	1	3	4
## 31	1	1	1	1	3
## 32	0	1	1	3	0
## 33	1	1	1	4	2
## 34	2	1	1	3	4
## 35	3	1	1	4	2

##	Result
## 1	home team win
## 2	away team win
## 3	home team win
## 4	away team win
## 5	home team win

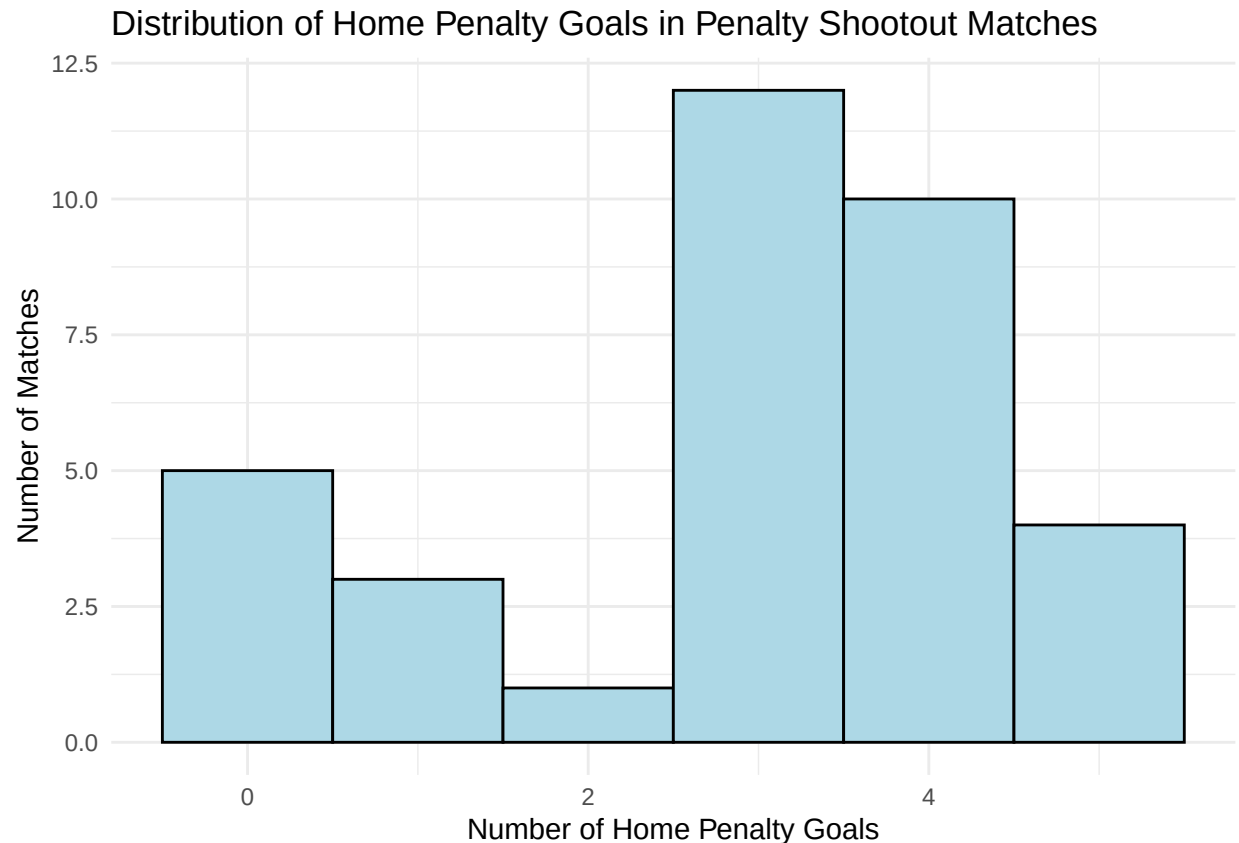
```
## 6  home team win
## 7  home team win
## 8  home team win
## 9  away team win
## 10 away team win
## 11 home team win
## 12 home team win
## 13 away team win
## 14 home team win
## 15 home team win
## 16 away team win
## 17 away team win
## 18 home team win
## 19 away team win
## 20 home team win
## 21 home team win
## 22 home team win
## 23 home team win
## 24 home team win
## 25 home team win
## 26 away team win
## 27 away team win
## 28 home team win
## 29 away team win
## 30 away team win
## 31 away team win
## 32 home team win
## 33 home team win
## 34 away team win
## 35 home team win
```

Since HomePenalty is a quantitative numeric variable, a histogram is selected for analyzing its distribution across matches involving a penalty shootout.

```
table(penalty_matches$HomePenalty)
```

```
##
##  0  1  2  3  4  5
##  5  3  1 12 10  4
```

```
library(ggplot2) # Use the ggplot2 library to produces the graphic with better customization
ggplot(penalty_matches, aes(x = HomePenalty)) +
  geom_histogram(binwidth = 1, fill = "lightblue", color = "black") + # Draw a histogram
  labs(title = "Distribution of Home Penalty Goals in Penalty Shootout Matches", # Add a title
       x = "Number of Home Penalty Goals", # Add label for x-axis
       y = "Number of Matches") + # Add label for y-axis
  theme_minimal() # Apply a minimal theme for a clean and simple appearance
```



Interpretation: This histogram indicates that most home teams score 3 or 4 penalty goals during the penalty shootouts. In contrast, lower scores (0-2 goals) and the maximum observed score (5 goals) occurred less often, suggesting that home teams generally centered around a moderate number of penalties during shootouts.

Question 2

TournamentName is stored in the tournaments dataset, while HomeTeamScore and Stage variables are stored in the matches dataset. Therefore, the two datasets are joined using TournamentID to analyse HomeTeamScore across different tournaments.

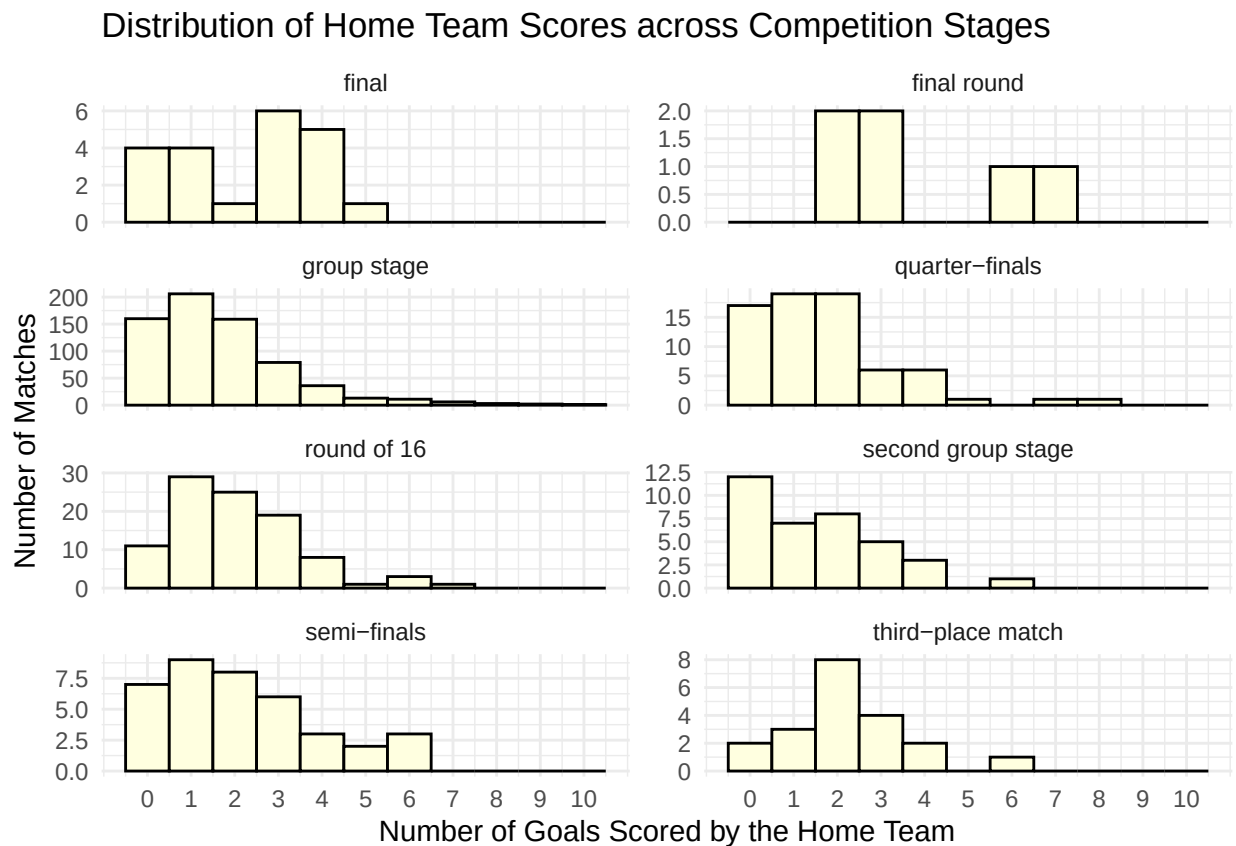
```
matches_tournaments <- matches %>%
  left_join(tournaments, by = "TournamentID") # Join TournamentName
str(matches_tournaments) # Validate if the TournamentName join successfully
```

```
## 'data.frame':   964 obs. of  17 variables:
## $ MatchID      : chr  "M-1930-01" "M-1930-02" "M-1930-03" "M-1930-04" ...
## $ TournamentID : chr  "WC-1930" "WC-1930" "WC-1930" "WC-1930" ...
## $ MatchName     : chr  "France v Mexico" "United States v Belgium" "Yugoslavia v Brazil" "Romania v" ...
## $ Stage         : Factor w/ 8 levels "final","final round",...: 3 3 3 3 3 3 3 3 3 ...
## $ MatchDate     : chr  "7/13/1930" "7/13/1930" "7/14/1930" "7/14/1930" ...
## $ MatchTime     : chr  "15:00:00" "15:00:00" "12:45:00" "14:50:00" ...
## $ StadiumID     : chr  "S-193" "S-192" "S-192" "S-193" ...
## $ HomeTeamID    : chr  "T-28" "T-80" "T-84" "T-59" ...
```

```
## $ AwayTeamID      : chr  "T-44" "T-06" "T-09" "T-54" ...
## $ HomeTeamScore   : int   4 3 2 3 1 3 4 3 1 1 ...
## $ AwayTeamScore   : int   1 0 1 1 0 0 0 0 0 0 ...
## $ ExtraTime       : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
## $ PenaltyShootout : int   0 0 0 0 0 0 0 0 0 0 ...
## $ HomePenalty      : num   0 0 0 0 0 0 0 0 0 0 ...
## $ AwayPenalty      : num   0 0 0 0 0 0 0 0 0 0 ...
## $ Result           : Factor w/ 3 levels "away team win",...: 3 3 3 3 3 3 3 3 3 3 ...
## $ TournamentName  : chr   "1930 FIFA World Cup" "1930 FIFA World Cup" "1930 FIFA World Cup" "1930 FIFA World Cup" ...
```

Create histogram to analyze the distribution of HomeTeamScore across different competition stages

```
ggplot(matches, aes(x = HomeTeamScore)) +
  geom_histogram(binwidth = 1, fill = "lightyellow", color = "black") + # Draw a histogram
  scale_x_continuous(breaks = seq(0, max(matches$HomeTeamScore), by = 1)) + # Display integer goals
  facet_wrap(~ Stage, scales = "free_y", ncol = 2) + # Compare across stages
  labs(title = "Distribution of Home Team Scores across Competition Stages", # Add a title
       x = "Number of Goals Scored by the Home Team", # Add label for x-axis
       y = "Number of Matches") + # Add label for y-axis
  theme_minimal() # Apply a minimal theme for a clean and simple appearance
```

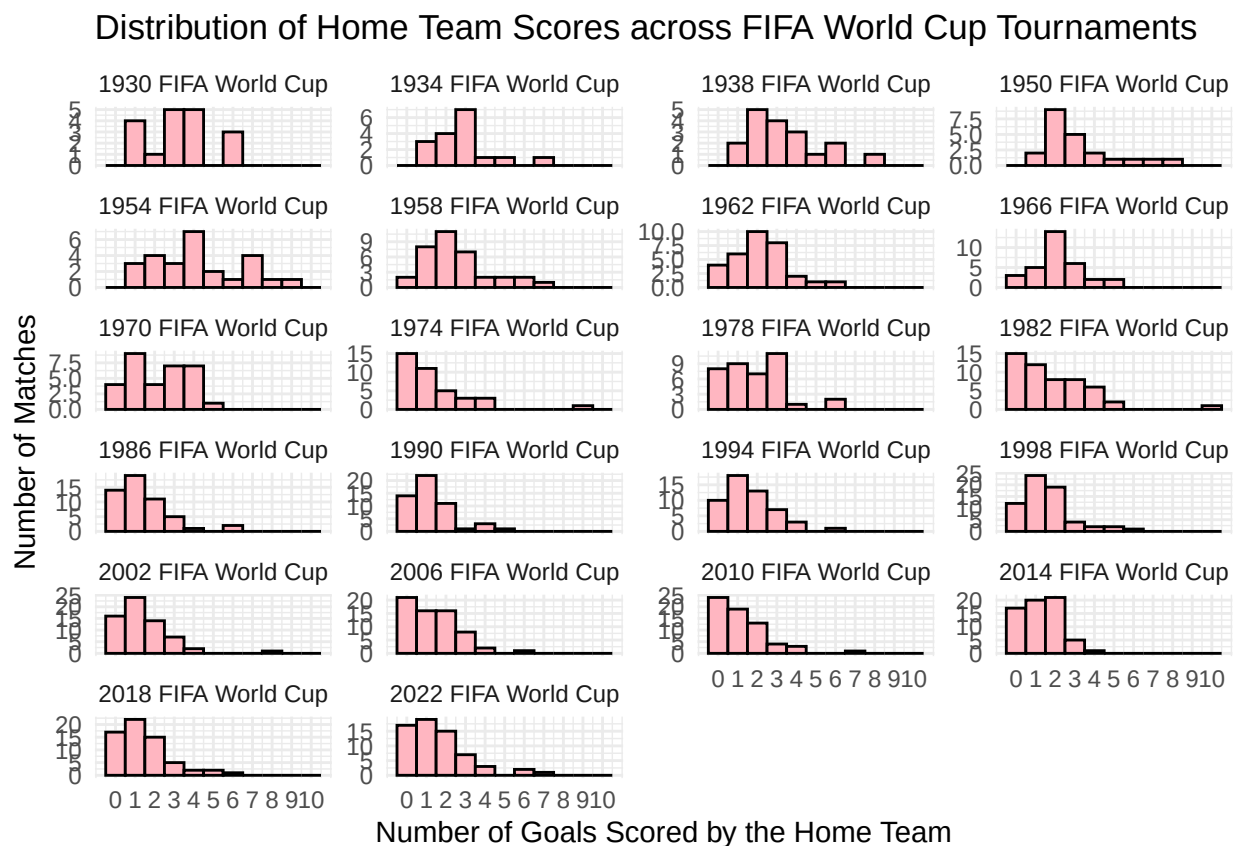


Interpretation: The overall distribution of home team scores across competition stages is mainly concentrated around 1-2 scores. Since the Group Stage contains the largest number of matches, it shows the

widespread of scores and a right-skewed distribution, indicating high-score performances from home team occurred occasionally. In contrast, the knockout stages, particularly final and final round, contain fewer matches and their distributions more concentrated at lower scores from the home teams. Additionally, across the remaining knockout stages, most home teams scored between 1 and 3 goals, while scoring 5 goals or more are relatively uncommon.

Create histogram to analyze the distribution of HomeTeamScore across different tournaments

```
ggplot(matches_tournaments, aes(x = HomeTeamScore)) +
  geom_histogram(binwidth = 1, fill = "lightpink", color = "black") + # Draw a histogram
  scale_x_continuous(breaks = seq(0, max(matches$HomeTeamScore), by = 1)) + # Display integer goals
  facet_wrap(~ TournamentName, scales = "free_y", ncol = 4) + # Compare across tournaments
  labs(title = "Distribution of Home Team Scores across FIFA World Cup Tournaments", # Add a title
       x = "Number of Goals Scored by the Home Team", # Add label for x-axis
       y = "Number of Matches") + # Add label for y-axis
  theme_minimal() # Apply a minimal theme for a clean and simple appearance
```



Interpretation: Overall, the distribution of home team scores across FIFA World Cup tournaments is concentrated between 0 and 3 goals, with most matches recording 1 to 2 goals. Additionally, later FIFA World Cup tournaments, particularly from 1998 onwards, contain more matches than earlier editions, resulting in taller histograms while maintaining similar home team scoring distribution. Although a few tournaments include matches where the home scored six or more goals, these high-scoring matches occurred only occasionally.

Create histogram to analyze the distribution of HomeTeamScore across AwayTeamScore groups

Categorize the AwayTeamScore across following groups: 0-1 goals, 2-3 goals, and 4 or more goals

```
matches <- matches %>%  
  mutate(AwayScoreGroup = case_when( # Add a new column based on conditions applied to AwayTeamScore  
    AwayTeamScore <= 1 ~ "0-1 goals", # Assign if the away team scored 0-1 goals  
    AwayTeamScore > 1 & AwayTeamScore <= 3 ~ "2-3 goals", # Assign if the away team scored 2-3 goals  
    AwayTeamScore > 3 ~ "4+ goals" # Assign if the away team scored more than 3 goals  
  ))
```

Convert AwayScoreGroup to an ordered factor for consistent group ordering in plots

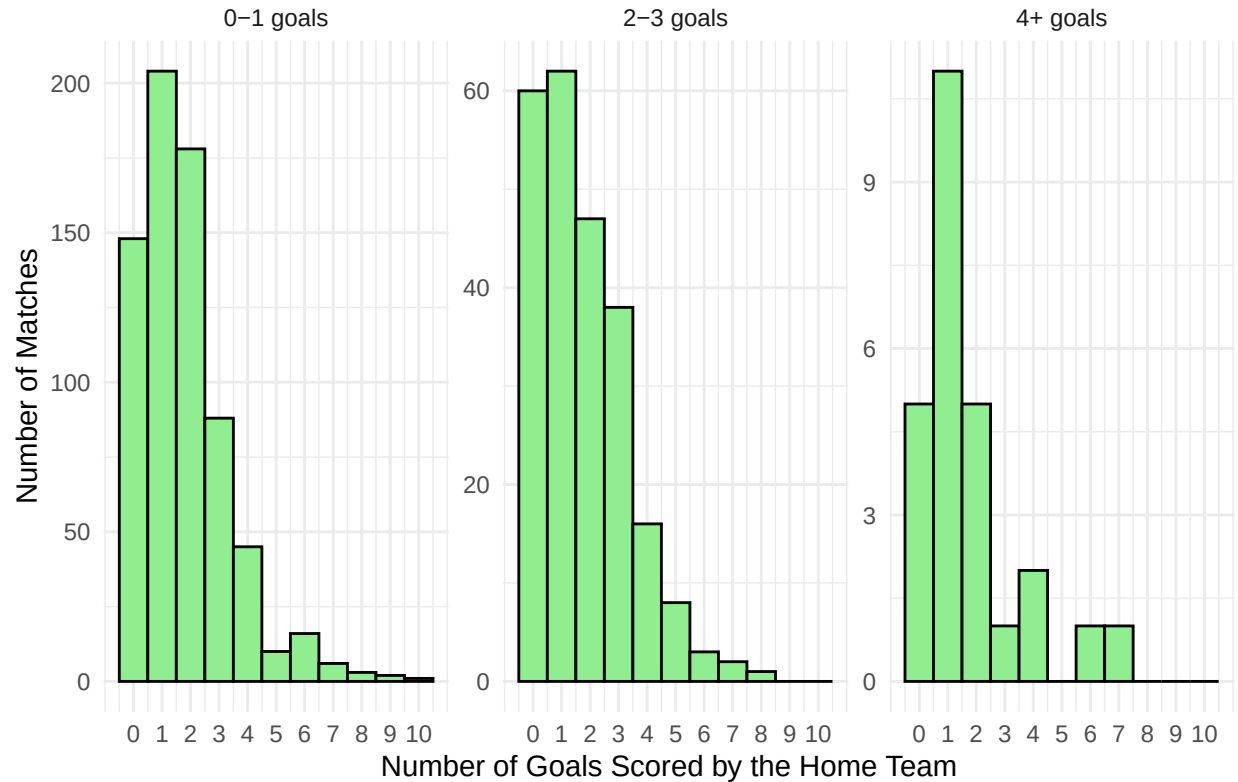
```
matches$AwayScoreGroup <- factor (  
  matches$AwayScoreGroup, # Specify the AwayScoreGroup variable to convert  
  levels = c("0-1 goals", "2-3 goals", "4+ goals") # Define the order of the factor levels  
)  
table(matches$AwayScoreGroup) # Validate the grouping of AwayScoreGroup before visualizing
```

```
##  
## 0-1 goals 2-3 goals 4+ goals  
##      701      237      26
```

Create histogram to analyze the distribution of HomeTeamScore across AwayTeamScore groups

```
ggplot(matches, aes(x = HomeTeamScore)) +  
  geom_histogram(binwidth = 1, fill = "lightgreen", color = "black") + # Draw a histogram  
  scale_x_continuous(breaks = seq(0, max(matches$HomeTeamScore), by = 1)) + # Display integer goals  
  facet_wrap(~ AwayScoreGroup, scales = "free_y") + # Compare across Away Team Score groups  
  labs(title = "Distribution of Home Team Scores by Away Team Score groups", # Add a title  
    x = "Number of Goals Scored by the Home Team", # Add label for x-axis  
    y = "Number of Matches") + # Add label for y-axis  
  theme_minimal() # Apply a minimal theme for a clean and simple appearance
```

Distribution of Home Team Scores by Away Team Score groups



Interpretation: The distribution of goals scored by Home Team is similar across the three Away Team Score groups, with most matches concentrated between 0 and 3 home goals regardless of the away team's scoring groups. From the distribution, while the 0-1 goals contains significant number of observations, the 4+ goals contains fewer observations, indicating that matches in which the away team scores 4 or more goals are relatively uncommon. Moreover, although there are few matches recorded five or more goals, these high-scoring goals are occasionally distributed across different groups of away team score.

Question 3

The matches with a penalty shootout have already been filtered out in Question 1

```
print(penalty_matches)
```

##	MatchID	TournamentID	MatchName	Stage
## 1	M-1982-50	WC-1982	West Germany v France	semi-finals
## 2	M-1986-45	WC-1986	Brazil v France	quarter-finals
## 3	M-1986-46	WC-1986	West Germany v Mexico	quarter-finals
## 4	M-1986-48	WC-1986	Spain v Belgium	quarter-finals
## 5	M-1990-41	WC-1990	Republic of Ireland v Romania	round of 16
## 6	M-1990-45	WC-1990	Argentina v Yugoslavia	quarter-finals
## 7	M-1990-49	WC-1990	Argentina v Italy	semi-finals
## 8	M-1990-50	WC-1990	West Germany v England	semi-finals
## 9	M-1994-44	WC-1994	Mexico v Bulgaria	round of 16
## 10	M-1994-48	WC-1994	Romania v Sweden	quarter-finals

## 11	M-1994-52	WC-1994		Brazil v Italy	final	
## 12	M-1998-56	WC-1998		Argentina v England	round of 16	
## 13	M-1998-57	WC-1998		Italy v France	quarter-finals	
## 14	M-1998-61	WC-1998		Brazil v Netherlands	semi-finals	
## 15	M-2002-52	WC-2002	Spain v Republic of Ireland		round of 16	
## 16	M-2002-59	WC-2002		Spain v South Korea	quarter-finals	
## 17	M-2006-54	WC-2006		Switzerland v Ukraine	round of 16	
## 18	M-2006-57	WC-2006		Germany v Argentina	quarter-finals	
## 19	M-2006-59	WC-2006		England v Portugal	quarter-finals	
## 20	M-2006-64	WC-2006		Italy v France	final	
## 21	M-2010-55	WC-2010		Paraguay v Japan	round of 16	
## 22	M-2010-58	WC-2010		Uruguay v Ghana	quarter-finals	
## 23	M-2014-49	WC-2014		Brazil v Chile	round of 16	
## 24	M-2014-52	WC-2014		Costa Rica v Greece	round of 16	
## 25	M-2014-60	WC-2014	Netherlands v Costa Rica		quarter-finals	
## 26	M-2014-62	WC-2014		Netherlands v Argentina	semi-finals	
## 27	M-2018-51	WC-2018		Spain v Russia	round of 16	
## 28	M-2018-52	WC-2018		Croatia v Denmark	round of 16	
## 29	M-2018-56	WC-2018		Colombia v England	round of 16	
## 30	M-2018-60	WC-2018		Russia v Croatia	quarter-finals	
## 31	M-2022-53	WC-2022		Japan v Croatia	round of 16	
## 32	M-2022-55	WC-2022		Morocco v Spain	round of 16	
## 33	M-2022-57	WC-2022		Croatia v Brazil	quarter-finals	
## 34	M-2022-58	WC-2022	Netherlands v Argentina		quarter-finals	
## 35	M-2022-64	WC-2022		Argentina v France	final	
##	MatchDate	MatchTime	StadiumID	HomeTeamID	AwayTeamID	HomeTeamScore
## 1	07/08/1982	21:00:00	S-159	T-83	T-28	3
## 2	6/21/1986	12:00:00	S-095	T-09	T-28	1
## 3	6/21/1986	16:00:00	S-102	T-83	T-44	0
## 4	6/22/1986	16:00:00	S-104	T-71	T-06	1
## 5	6/25/1990	17:00:00	S-073	T-58	T-59	0
## 6	6/30/1990	17:00:00	S-071	T-03	T-84	0
## 7	07/03/1990	20:00:00	S-076	T-03	T-39	1
## 8	07/04/1990	20:00:00	S-082	T-83	T-27	1
## 9	07/05/1994	16:30:00	S-184	T-44	T-10	1
## 10	07/10/1994	12:30:00	S-189	T-59	T-72	2
## 11	7/17/1994	12:30:00	S-187	T-09	T-39	0
## 12	6/30/1998	21:00:00	S-049	T-03	T-27	2
## 13	07/03/1998	16:30:00	S-048	T-39	T-28	0
## 14	07/07/1998	21:00:00	S-042	T-09	T-46	1
## 15	6/16/2002	20:30:00	S-145	T-71	T-58	1
## 16	6/22/2002	15:30:00	S-140	T-71	T-69	0
## 17	6/26/2006	21:00:00	S-054	T-73	T-78	0
## 18	6/30/2006	17:00:00	S-053	T-29	T-03	1
## 19	07/01/2006	17:00:00	S-058	T-27	T-56	0
## 20	07/09/2006	20:00:00	S-053	T-39	T-28	1
## 21	6/29/2010	16:00:00	S-135	T-53	T-42	0
## 22	07/02/2010	20:30:00	S-131	T-81	T-30	1
## 23	6/28/2014	13:00:00	S-008	T-09	T-13	1
## 24	6/29/2014	17:00:00	S-018	T-16	T-31	1
## 25	07/05/2014	17:00:00	S-021	T-46	T-16	0
## 26	07/09/2014	17:00:00	S-022	T-46	T-03	0
## 27	07/01/2018	17:00:00	S-117	T-71	T-60	1
## 28	07/01/2018	21:00:00	S-119	T-17	T-21	1

## 29	07/03/2018	21:00:00	S-118	T-15	T-27	1
## 30	07/07/2018	21:00:00	S-124	T-60	T-17	2
## 31	12/05/2022	18:00:00	S-111	T-42	T-17	1
## 32	12/06/2022	18:00:00	S-109	T-45	T-71	0
## 33	12/09/2022	18:00:00	S-109	T-17	T-09	1
## 34	12/09/2022	22:00:00	S-114	T-46	T-03	2
## 35	12/18/2022	18:00:00	S-114	T-03	T-28	3

##	AwayTeamScore	ExtraTime	PenaltyShootout	HomePenalty	AwayPenalty
## 1	3	1	1	0	4
## 2	1	1	1	0	4
## 3	0	1	1	0	1
## 4	1	1	1	0	5
## 5	0	1	1	5	4
## 6	0	1	1	3	2
## 7	1	1	1	4	3
## 8	1	1	1	4	3
## 9	1	1	1	1	3
## 10	2	1	1	4	5
## 11	0	1	1	3	2
## 12	2	1	1	4	3
## 13	0	1	1	3	4
## 14	1	1	1	4	2
## 15	1	1	1	3	2
## 16	0	1	1	3	5
## 17	0	1	1	0	3
## 18	1	1	1	4	2
## 19	0	1	1	1	3
## 20	1	1	1	5	3
## 21	0	1	1	5	3
## 22	1	1	1	4	2
## 23	1	1	1	3	2
## 24	1	1	1	5	3
## 25	0	1	1	4	3
## 26	0	1	1	2	4
## 27	1	1	1	3	4
## 28	1	1	1	3	2
## 29	1	1	1	3	4
## 30	2	1	1	3	4
## 31	1	1	1	1	3
## 32	0	1	1	3	0
## 33	1	1	1	4	2
## 34	2	1	1	3	4
## 35	3	1	1	4	2

##	Result
## 1	home team win
## 2	away team win
## 3	home team win
## 4	away team win
## 5	home team win
## 6	home team win
## 7	home team win
## 8	home team win
## 9	away team win
## 10	away team win

```
## 11 home team win
## 12 home team win
## 13 away team win
## 14 home team win
## 15 home team win
## 16 away team win
## 17 away team win
## 18 home team win
## 19 away team win
## 20 home team win
## 21 home team win
## 22 home team win
## 23 home team win
## 24 home team win
## 25 home team win
## 26 away team win
## 27 away team win
## 28 home team win
## 29 away team win
## 30 away team win
## 31 away team win
## 32 home team win
## 33 home team win
## 34 away team win
## 35 home team win
```

Top 5 Home Teams in Penalty Shootouts

```
topHomePenalty <- penalty_matches %>% # Count the number of matches each home team participated in
  group_by(HomeTeamID) %>% # Group all matches by the HomeTeamID
  summarize(NumberOfMatches = n(), .groups = "drop") %>% # Count the number of matches
  left_join(teams, by = c("HomeTeamID" = "TeamID")) %>% # Join the two datasets
  select(TeamName, TeamCode, NumberOfMatches) %>% # Keep only required columns
  arrange(desc(NumberOfMatches)) %>% # Sort teams from the highest to the lowest number of matches
  slice_head(n = 5) # Return only the top 5 home teams
topHomePenalty # Display the top 5 home teams in penalty shootouts
```

```
## # A tibble: 5 x 3
##   TeamName      TeamCode NumberOfMatches
##   <chr>         <chr>          <int>
## 1 Argentina    ARG              4
## 2 Brazil       BRA              4
## 3 Spain        ESP              4
## 4 Netherlands  NLD              3
## 5 West Germany DEU              3
```

Top 5 Away Teams in Penalty Shootouts

```
topAwayPenalty <- penalty_matches %>% # Count the number of matches each away team participated in
  group_by(AwayTeamID) %>% # Group all matches by the AwayTeamID
```

```

summarize(NumberOfMatches = n(), .groups = "drop") %>% # Count the number of matches
left_join(teams, by = c("AwayTeamID" = "TeamID")) %>% # Join the two datasets
select(TeamName, TeamCode, NumberOfMatches) %>% # Keep only required columns
arrange(desc(NumberOfMatches)) %>% # Sort teams from the highest to the lowest number of matches
slice_head(n = 5) # Return only the top 5 away teams
topAwayPenalty # Display the top 5 away teams in penalty shootouts

```

```

## # A tibble: 5 x 3
##   TeamName TeamCode NumberOfMatches
##   <chr>     <chr>           <int>
## 1 France   FRA                 5
## 2 Argentina ARG                 3
## 3 England  ENG                 3
## 4 Croatia  HRV                 2
## 5 Italy     ITA                 2

```

Discussion: The results indicate that Argentina, Brazil, Spain, the Netherlands, and West Germany were the top 5 home teams participated in FIFA World Cup matches decided by penalty shootouts. On the away team side, France, Argentina, England, Croatia, and Italy were the top five. There is different between the rankings of home teams and away teams, with only Argentina appearing on both lists. Viewing the two lists, France recorded the highest number of participation of an away team in penalty matches with 5 matches, while Argentina, Brazil, and Spain shared the highest number of home appearance with each participate 4 matches. Therefore, the top national teams participating in matches with penalty shootouts were not identical when home and away matches were considered separately.

Question 4

While there are many factors influence the match outcome (Result) in the FIFA World Cup, this assignment will examine the influence of Stage and ExtraTime for the following analysis.

Factor 1: Stage

Create table to present the influence of Stage on Result

```

stage_result_table <- table(matches$Stage, matches$Result) # Create frequency table
stage_result_table # View the table

```

```

##
##           away team win draw home team win
##   final                4    0             17
##   final round           0    1             5
##   group stage          164  166            346
##   quarter-finals       24    2             44
##   round of 16           21    2             74
##   second group stage    11    8             17
##   semi-finals           10    0             28
##   third-place match      6    0             14

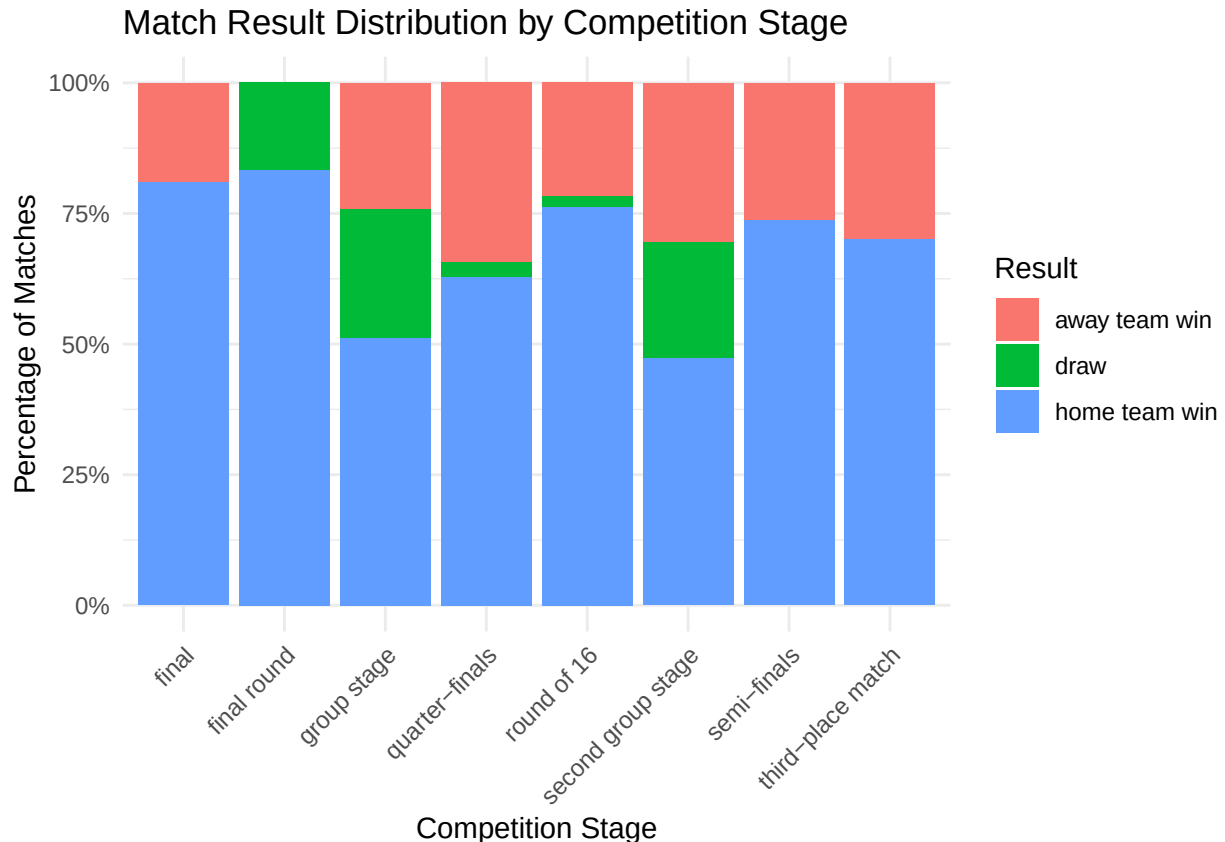
```

```
round(prop.table(stage_result_table, 1) * 100, 1) # View the table with percentage
```

```
##
##           away team win draw home team win
## final           19.0  0.0           81.0
## final round           0.0 16.7           83.3
## group stage          24.3 24.6           51.2
## quarter-finals        34.3  2.9           62.9
## round of 16           21.6  2.1           76.3
## second group stage     30.6 22.2           47.2
## semi-finals           26.3  0.0           73.7
## third-place match      30.0  0.0           70.0
```

Create a proportional stacked bar chart to compare match results across stages

```
ggplot(matches, aes(x = Stage, fill = Result)) +
  geom_bar(position = "fill") + # Draw stacked bars scaled to proportions (each bar sums to 100%)
  scale_y_continuous(labels = scales::percent) + # Display the y-axis as percentages
  labs(title = "Match Result Distribution by Competition Stage", # Add a title
       x = "Competition Stage", # Add label for x-axis
       y = "Percentage of Matches", # Add label for y-axis
       fill = "Result") + # Set the title of the legend
  theme_minimal() + # Apply a minimal theme for a clean and simple appearance
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) # Rotate the x-axis labels
```



Discussion on findings: The distribution of match result varies across different competition stages. Home team win accounts for largest proportion of results in every stage. The final round and the Round of 16 show the highest proportion of home team victories, with home team winning more than third quarters of the number of matches. In contrast, the Group Stage and Second Group Stage have a noticeably higher proportions of draws more than knockout stages, indicating that draw matches are commonly occur in the beginning of the tournament. Away team has a consistent proportion across stages, accounting for around one-quarter to one-third of matches in a tournament. To conclude, competition stage is associated with changes of match outcomes with a reduced frequency of draws in knockout rounds.

Factor 2: Extra Time

Create table to present the influence of Extra Time on Result

```
extratime_result_table <- table(matches$ExtraTime, matches$Result) # Create frequency table
extratime_result_table # View the table
```

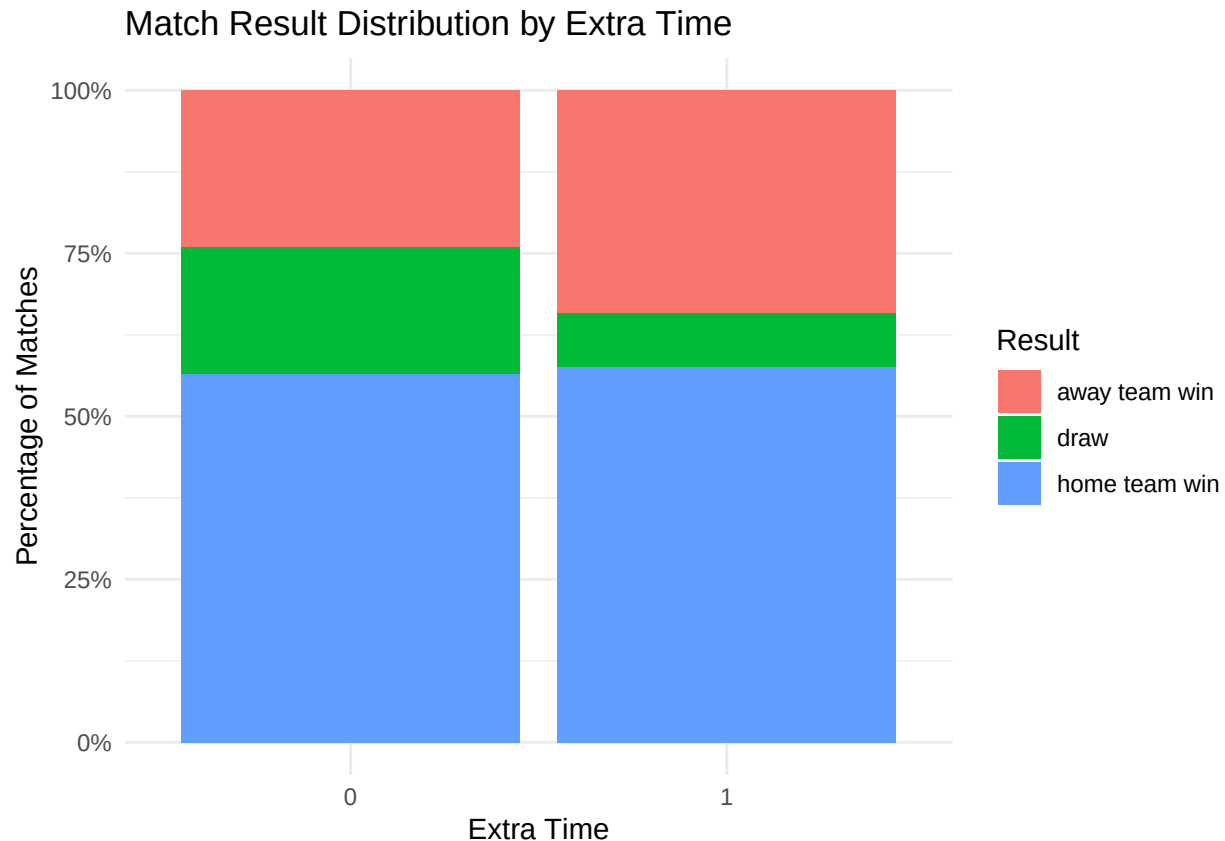
```
##
##      away team win draw home team win
##    0              215  173             503
##    1              25   6              42
```

```
round(prop.table(extratime_result_table, 1) * 100, 1) # View the table with percentage to support
```

```
##
##      away team win draw home team win
##    0              24.1 19.4             56.5
##    1              34.2  8.2             57.5
```

Create a proportional stacked bar chart to compare match results by Extra Time

```
ggplot(matches, aes(x = ExtraTime, fill = Result)) +
  geom_bar(position = "fill") + # Draw stacked bars scaled to proportions (each bar sums to 100%)
  scale_y_continuous(labels = scales::percent) + # Display the y-axis as percentage
  labs(title = "Match Result Distribution by Extra Time", # Add a title
       x = "Extra Time", # Add label for x-axis
       y = "Percentage of Matches", # Add label for y-axis
       fill = "Result") + # Set the title of the legend
  theme_minimal() # Apply a minimal theme for a clean and simple appearance
```



Discussion on findings: The distribution of match outcomes has slightly difference depending on the match proceed extra time or not. Matches with extra time recorded a lower proportion of draw and a higher proportion of away team wins compared with matches within regular time. Additionally, the proportion of home team win is generally similar between two groups, making up more than half of all matches in both case. In conclusion, the difference is modest, indicating that while extra time influences match results home team victories remain nearly the same whether extra time is proceeded.

Question 5

Create a GitHub repository with all relevant files supporting this analysis

The GitHub repository link: https://github.com/hieuvantnt-del/AP1_Assignment-2_TRANNgocHieuVan_22205175

History of creating git repository and committing each question (screenshot)

Initial Git setup on Terminal

```
[tranngochieuvan@Annies-MacBook-Pro AP1_Assignment2 % pwd
/Users/tranngochieuvan/AP1_Assignment2
[tranngochieuvan@Annies-MacBook-Pro AP1_Assignment2 % ls
A2_RScript.R    Matches.csv    Stadiums.csv   Teams.csv      Tournaments.csv
```

pwd: Display the current working directory.

ls: List all files in the current directory.

```
[tranngochieuvan@Annies-MacBook-Pro AP1_Assignment2 % git init ]
Initialized empty Git repository in /Users/tranngochieuvan/AP1_Assignment2/.git/
git init: Initialise a new Git repository.

[tranngochieuvan@Annies-MacBook-Pro AP1_Assignment2 % git add . ]
[tranngochieuvan@Annies-MacBook-Pro AP1_Assignment2 % git status ]
On branch main

No commits yet
```

```
Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   A2_RScript.R
    new file:   Matches.csv
    new file:   Stadiums.csv
    new file:   Teams.csv
    new file:   Tournaments.csv
```

git add.: Stage all project files for the initial commit.

git status: Check the repository status after modifying files.

Initial commit of all the files in the working directory

Staged all the modified files in the working directory and create initial commit git commit -m: Commit the initial datasets and Rscript file with.

Connect the local repository to the GitHub repository

```
[tranngochieuvan@Annies-MacBook-Pro AP1_Assignment2 % git remote -v ]
[tranngochieuvan@Annies-MacBook-Pro AP1_Assignment2 % git remote add origin git@github.com:hieuvantnt-del/AP1_Assignment-2_TRANNgocHieuVan_22205175.git ]
[tranngochieuvan@Annies-MacBook-Pro AP1_Assignment2 % git branch -M main ]
[tranngochieuvan@Annies-MacBook-Pro AP1_Assignment2 % git push -u origin main ]
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 10 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (7/7), 21.99 KiB | 4.40 MiB/s, done.
Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:hieuvantnt-del/AP1_Assignment-2_TRANNgocHieuVan_22205175.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

git remote -v: Show the configured remote repositories.

git remote and origin: Connect the local repository to the GitHub repository.

git branch -M main: Rename the default branch to main.

git push -u origin main: Push the local repository to GitHub and set the upstream branch.

Question 1

```
[tranngochieuvan@Annies-MacBook-Pro AP1_Assignment2 % git add A2_RScript.R ]
[tranngochieuvan@Annies-MacBook-Pro AP1_Assignment2 % git commit -m "Complete Question 1 Analysis"
[main 2a76445] Complete Question 1 Analysis
 1 file changed, 69 insertions(+)
[tranngochieuvan@Annies-MacBook-Pro AP1_Assignment2 % git push ]
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 10 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.34 KiB | 1.34 MiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:hieuvantnt-del/AP1_Assignment-2_TRANNgocHieuVan_22205175.git
 467f255..2a76445  main -> main
```

git add: Stage the A2_RScript.R file for the second commit.

git commit -m: Commit the the completed Question 1.

git push: Push the second commit to GitHub.

Question 2

```
[tranngochieuvan@Annies-MacBook-Pro AP1_Assignment2 % git add A2_RScript.R ]
[tranngochieuvan@Annies-MacBook-Pro AP1_Assignment2 % git commit -m "Complete Question visualization"
[main af65ff0] Complete Question visualization
 1 file changed, 68 insertions(+), 1 deletion(-)
[tranngochieuvan@Annies-MacBook-Pro AP1_Assignment2 % git push ]
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 10 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 2.51 KiB | 2.51 MiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:hieuvantnt-del/AP1_Assignment-2_TRANNgocHieuVan_22205175.git
 2a76445..af65ff0  main -> main
```

git add: Stage the A2_RScript.R file for the third commit.

git commit -m: Commit the the completed Question 2.

git push: Push the third commit to GitHub.

Question 3

```
[tranngochieuvan@Annies-MacBook-Pro AP1_Assignment2 % git add A2_RScript.R ]
[tranngochieuvan@Annies-MacBook-Pro AP1_Assignment2 % git commit -m "Complete Question 3 analysis"
[main c4601e5] Complete Question 3 analysis
 1 file changed, 33 insertions(+), 1 deletion(-)
[tranngochieuvan@Annies-MacBook-Pro AP1_Assignment2 % git push ]
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 10 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.17 KiB | 1.17 MiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:hieuvantnt-del/AP1_Assignment-2_TRANNgocHieuVan_22205175.git
   af65ff0..c4601e5  main -> main
```

git add: Stage the A2_RScript.R file for the fourth commit.

git commit -m: Commit the the completed Question 2.

git push: Push the fourth commit to GitHub.

Question 4

```
[tranngochieuvan@Annies-MacBook-Pro AP1_Assignment2 % git add A2_RScript.R ]
[tranngochieuvan@Annies-MacBook-Pro AP1_Assignment2 % git commit -m "Complete Question 4 visualization and analysis"
[main 2a2df06] Complete Question 4 visualization and analysis
 1 file changed, 45 insertions(+)
[tranngochieuvan@Annies-MacBook-Pro AP1_Assignment2 % git push ]
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 10 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.53 KiB | 1.53 MiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:hieuvantnt-del/AP1_Assignment-2_TRANNgocHieuVan_22205175.git
   64f38e8..2a2df06  main -> main
```

git add: Stage the A2_RScript.R file for the fifth commit.

git commit -m: Commit the the completed Question 2.

git push: Push the fifth commit to GitHub.